Name – Aarush Khullar

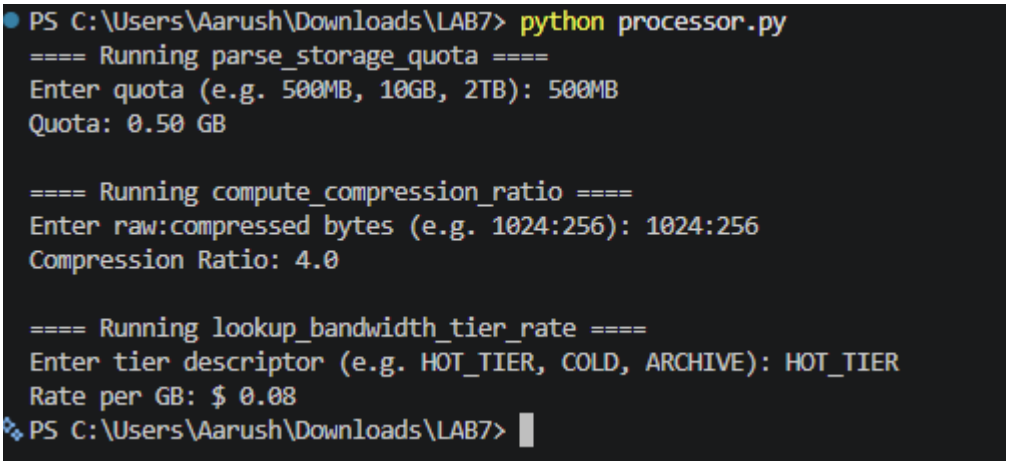
Section – 5A

SRN – PES2UG24CS014

SE lab 07

Repo link – https://github.com/Aarush-Khullar/SE-LAB--07

Screenshot 1 – before fix

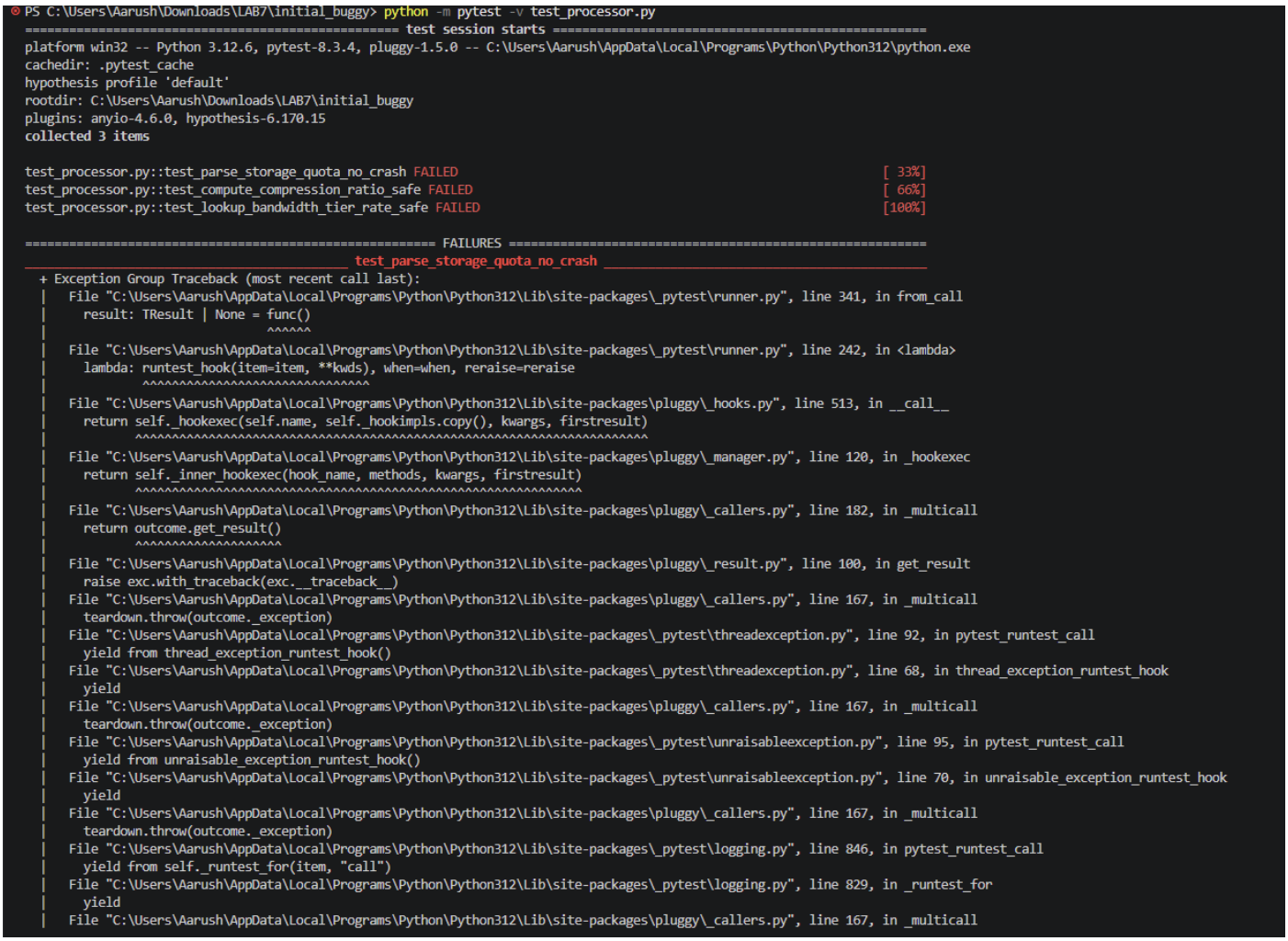


```
PS C:\Users\Aarush\Downloads\LAB7> python processor.py
==== Running parse_storage_quota ====
Enter quota (e.g. 500MB, 10GB, 2TB): 500MB
Quota: 0.50 GB

==== Running compute_compression_ratio ====
Enter raw:compressed bytes (e.g. 1024:256): 1024:256
Compression Ratio: 4.0

==== Running lookup_bandwidth_tier_rate ====
Enter tier descriptor (e.g. HOT_TIER, COLD, ARCHIVE): HOT_TIER
Rate per GB: $ 0.08
PS C:\Users\Aarush\Downloads\LAB7>
```

SCREENSHOT 2 – Test file before fix



```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\Aarush\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aarush\Downloads\LAB7\initial_buggy
plugins: anyio-4.6.0, hypothesis-6.170.15
collected 3 items

test_processor.py::test_parse_storage_quota_no_crash FAILED              [ 33%]
test_processor.py::test_compute_compression_ratio_safe FAILED            [ 66%]
test_processor.py::test_lookup_bandwidth_tier_rate_safe FAILED           [100%]

================================== FAILURES ===================================
_____________________________ test_parse_storage_quota_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |            ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |            ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 10, in test_parse_storage_quota_no_crash
  |     def test_parse_storage_quota_no_crash(s):
  |                ^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 12, in test_parse_storage_quota_no_crash
    |     parse_storage_quota(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 15, in parse_storage_quota
    |     gb_total = value * multipliers[unit]
    |                        ~~~~~~~~~~~^^^^^^
    | KeyError: '00'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 14, in test_parse_storage_quota_no_crash
    |     assert False, f"parse_storage_quota crashed with input {s!r}: {e}"
    | AssertionError: parse_storage_quota crashed with input '000': '00'
    | assert False
    | Failing test case: test_parse_storage_quota_no_crash(
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
    | assert False
    | Failing test case: test_parse_storage_quota_no_crash(
    |     s='000',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 12, in test_parse_storage_quota_no_crash
    |     parse_storage_quota(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 9, in parse_storage_quota
    |     value = int(val_str)
    |             ^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 14, in test_parse_storage_quota_no_crash
    |     assert False, f"parse_storage_quota crashed with input {s!r}: {e}"
    | AssertionError: parse_storage_quota crashed with input '': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_storage_quota_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 12, in test_parse_storage_quota_no_crash
    |     parse_storage_quota(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 6, in parse_storage_quota
    |     cleaned = quota_str.strip().upper()
    |               ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 14, in test_parse_storage_quota_no_crash
    |     assert False, f"parse_storage_quota crashed with input {s!r}: {e}"
    | AssertionError: parse_storage_quota crashed with input None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_parse_storage_quota_no_crash(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
_____________________________ test_compute_compression_ratio_safe ______________________________
  + Exception Group Traceback (most recent call last):
    |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
    |     result: TResult | None = func()
    |                             ^^^^^^
    |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
    |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
    |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
  |           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 18, in test_compute_compression_ratio_safe
  |     def test_compute_compression_ratio_safe(s):
  |                    ^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 20, in test_compute_compression_ratio_safe
    |     compute_compression_ratio(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 24, in compute_compression_ratio
    |     raw_bytes = float(raw_str)
    |                 ^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 22, in test_compute_compression_ratio_safe
    |     assert False, f"compute_compression_ratio crashed with input {s!r}: {e}"
    | AssertionError: compute_compression_ratio crashed with input ':': could not convert string to float: ''
    | assert False
    | Failing test case: test_compute_compression_ratio_safe(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 20, in test_compute_compression_ratio_safe
    |     compute_compression_ratio(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 23, in compute_compression_ratio
    |     raw_str, comp_str = size_metric_str.split(':')
    |     ^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 22, in test_compute_compression_ratio_safe
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 22, in test_compute_compression_ratio_safe
    |     assert False, f"compute_compression_ratio crashed with input {s!r}: {e}"
    | AssertionError: compute_compression_ratio crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_compute_compression_ratio_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 20, in test_compute_compression_ratio_safe
    |     compute_compression_ratio(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 23, in compute_compression_ratio
    |     raw_str, comp_str = size_metric_str.split(':')
    |                         ^^^^^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 22, in test_compute_compression_ratio_safe
    |     assert False, f"compute_compression_ratio crashed with input {s!r}: {e}"
    | AssertionError: compute_compression_ratio crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_compute_compression_ratio_safe(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
_____________________________ test_lookup_bandwidth_tier_rate_safe _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                              ^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 26, in test_lookup_bandwidth_tier_rate_safe
  |     def test_lookup_bandwidth_tier_rate_safe(s):
  |                    ^^^
  |   File "C:\Users\Aarush\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 28, in test_lookup_bandwidth_tier_rate_safe
```

```
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy> python -m pytest -v test_processor.py
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 28, in test_lookup_bandwidth_tier_rate_safe
    |     lookup_bandwidth_tier_rate(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 40, in lookup_bandwidth_tier_rate
    |     return tier_rates[prefix]
    |            ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 30, in test_lookup_bandwidth_tier_rate_safe
    |     assert False, f"lookup_bandwidth_tier_rate crashed with input {s!r}: {e}"
    | AssertionError: lookup_bandwidth_tier_rate crashed with input '': ''
    | assert False
    | Failing test case: test_lookup_bandwidth_tier_rate_safe(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py:29
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 28, in test_lookup_bandwidth_tier_rate_safe
    |     lookup_bandwidth_tier_rate(s)
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\processor.py", line 34, in lookup_bandwidth_tier_rate
    |     prefix = tier_descriptor.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py", line 30, in test_lookup_bandwidth_tier_rate_safe
    |     assert False, f"lookup_bandwidth_tier_rate crashed with input {s!r}: {e}"
    | AssertionError: lookup_bandwidth_tier_rate crashed with input None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_bandwidth_tier_rate_safe(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Aarush\Downloads\LAB7\initial_buggy\test_processor.py:29
    +------------------------------------
=========================================== short test summary info ============================================
FAILED test_processor.py::test_parse_storage_quota_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_compute_compression_ratio_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_lookup_bandwidth_tier_rate_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================ 3 failed in 5.28s =============================================
PS C:\Users\Aarush\Downloads\LAB7\initial_buggy>
```

Screenshot 3 – After fix

```
PS C:\Users\Aarush\Downloads\LAB7> python processor.py
==== Running parse_storage_quota ====
Enter quota (e.g. 500MB, 10GB, 2TB): 500MB
Quota: 0.50 GB

==== Running compute_compression_ratio ====
Enter raw:compressed bytes (e.g. 1024:256): 1024:256
Compression Ratio: 4.0

==== Running lookup_bandwidth_tier_rate ====
Enter tier descriptor (e.g. HOT_TIER, COLD, ARCHIVE): HOT_TIER
Rate per GB: $ 0.08
PS C:\Users\Aarush\Downloads\LAB7> python -m pytest -v test_processor.py
==================================== test session starts ====================================
platform win32 -- Python 3.12.6, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\Aarush\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Aarush\Downloads\LAB7
plugins: anyio-4.6.0, hypothesis-6.170.15
collected 3 items

test_processor.py::test_parse_storage_quota_no_crash PASSED                              [ 33%]
test_processor.py::test_compute_compression_ratio_safe PASSED                            [ 66%]
test_processor.py::test_lookup_bandwidth_tier_rate_safe PASSED                           [100%]

==================================== 3 passed in 3.03s =====================================
PS C:\Users\Aarush\Downloads\LAB7>
```